# Romina Saljooghian

Automation & AI Engineer

## CONTACT


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## TECHNICAL SKILLS

**Programming**
Python, MATLAB, SQL

**ML & Deep Learning**
PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face

**LLM & Agentic AI**
LangGraph, LangChain, OpenAI API, Anthropic API, Prompt Engineering

**Backend & API**
FastAPI, REST API, Pydantic, pytest

**MLOps & Tools**
MLflow, Hydra, Git, Linux, LaTeX

**Control & Robotics**
Simulink, MPC, LQR, Kalman Filter, PID, Kinematics, System ID

## LANGUAGES

**English** — Fluent
**Italian** — A2
**Persian** — Native

## CERTIFICATIONS

**LangGraph Essentials**
LangChain Academy, 2026

**Automation Explorer**
UiPath Academy, 2026

**Python**
freeCodeCamp, 2026

**Relational Databases**
freeCodeCamp, 2026

## INTERESTS

Piano · Puzzles

## PROFILE

MSc graduate in Automation and Control Engineering with experience in deep learning, computer vision, agentic AI, and control systems. Built end-to-end ML pipelines from foundation-model orchestration to production REST APIs with LangGraph. Seeking roles in ML Engineering, CV, or Robotics AI.

## EDUCATION

**M.Sc. Automation & Control Engineering** — Sep 2022 – Mar 2026
*Politecnico di Milano, Italy* — *107/110*
Thesis: Open-vocabulary scene graph generation for robotic manipulation using foundation models (CLIP, SAM, Grounding DINO, RAM) — MERLIN Lab.

**B.Sc. Electrical Engineering** (Minor: Control) — Sep 2016 – Aug 2020
*K.N.Toosi University of Technology, Tehran* — *16.16/20*

**Selective Coursework:** Neural Networks & Deep Learning · Control of Industrial Robots · Multiagent Systems · Software Engineering · Model Identification · Data-Driven Control · Autonomous Vehicles

## PROJECTS

**Scene Graph Generation for Robotic Manipulation** — 2025 – 2026
*MSc Thesis · Python, PyTorch, CLIP, SAM, GPT API, Gemini API, Hydra*
- Designed a modular open-vocabulary pipeline chaining 4 foundation models; **6× improvement** over end-to-end baseline (PredCls mR@100: 48.90%)
- Built VLM-based annotation pipeline (GPT + Gemini) with structured prompts, producing 246 spatial-relationship triplets for a novel robotics dataset
- Zero-shot cross-domain transfer to manipulation scenes (SGDet mR@20: 16.13%) with no fine-tuning

**Visual QA Agent** — 2026
*Personal Project · Python, FastAPI, LangGraph, Anthropic API, OpenAI API, PyMuPDF*
- Built agentic backend answering visual questions about images and PDFs through a **4-node LangGraph pipeline** with conditional error routing at every stage
- Dual-provider VLM abstraction (Claude + GPT-4o); PDF-to-image conversion via PyMuPDF; 11 automated tests with full coverage

**Plant Leaf Disease Classification** — 2024
*Course Project · Python, TensorFlow, FastAPI, MLflow, pytest*
- 5-model ensemble CNN achieving **86% accuracy**; async REST API with Grad-CAM explainability endpoint
- MLflow experiment tracking with nested runs; 6-module pytest suite covering API, data pipeline, and model registry

**Time-Series Forecasting** — 2024
*Course Competition · Python, TensorFlow, Keras*
- Benchmarked 6+ architectures on 48K time series; Conv. LSTM with autoregressive decoding achieved **test MSE of 0.01**

**CNC Machine — 2-DOF Pantograph Robot** — 2024
*Lab Project · MATLAB, Simulink, Quanser SRV02 Hardware*
- Deployed 9 controllers (PID, LQR, MPC, Kalman) on real hardware; MPC: **0.26 s settling, <0.5% overshoot**
- Near-perfect 2D trajectory tracking with KF + LQR; validated in both time and frequency domains on physical plant

**Networked Control of a Vehicle Platoon** — 2025
*Course Project · MATLAB*
- LMI-based state-feedback controllers for a 4-vehicle platoon under 6 communication topologies; benchmarked centralized vs. distributed cost and convergence trade-offs

## WORK EXPERIENCE

**Software Developer** — Jan 2020 – Sep 2021
*Torfeh Negar Holding, Tehran, Iran*
- Built production UI and backend modules for Holoo accounting software serving thousands of users
- Optimized SQL Server queries; resolved bottlenecks and stabilized nightly batch processes

**Instrumentation & Control Intern** — Jun – Sep 2019
*Tavan Gostare Zino, Tehran, Iran*
- Designed and tested industrial grounding and earthing systems for IEC standards compliance